

Smart Recycling Helper

Classifying waste from a single photo

Tier 1: Image Classification

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ITAI 1378 — Midterm Blueprint



The Problem

A simple tool for identifying common waste categories.

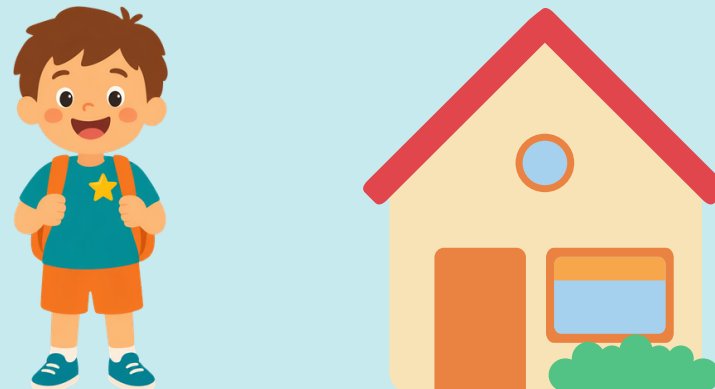
What is the problem?

People may be unsure which waste category an item belongs to.



Who could use it?

Students, households, schools, and businesses sorting everyday waste.



Why does it matter?

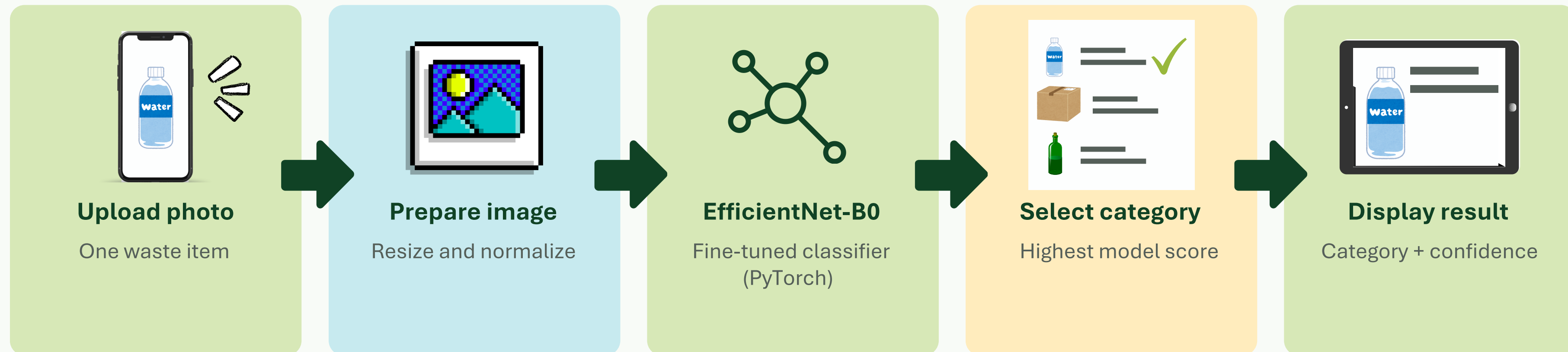
Identifying the category is a useful first step before checking local disposal rules.



Our focus: identify one item in one photo.

Our Solution Workflow

Our system will classify a photo of one waste item and display its predicted category with a confidence score.



Planned output categories

Cardboard • Glass • Metal • Paper • Plastic • Trash (Thung and Yang)

Predicts a category, users check local disposal rules.

4 Technical Approach

 Smart Recycling

We will use transfer learning to adapt EfficientNet-B0 to classify waste images.

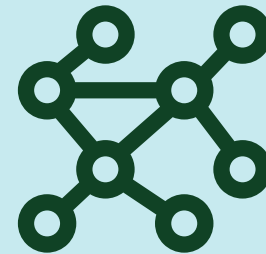
CV Technique



Image Classification

Predict one waste category for each uploaded photo.

Architecture & Model



Convolutional Neural Network (CNN)

EfficientNet-B0 (PyTorch)

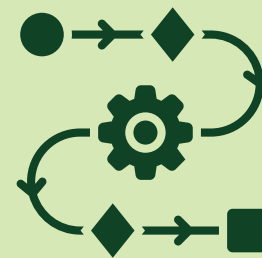
How We Will Use It



Transfer Learning

Start with pretrained weights, replace the final classifier with six outputs, and train it using TrashNet.

Framework



PyTorch + Torchvision

Load the model, train it, and generate predictions in Google Colab.

Why This Approach?



Transfer learning gives us a starting point instead of training a model from scratch. One classifier keeps the project focused and fits our Tier 1 scope.



5 Data Plan

Understanding the dataset used to train and evaluate our classifier

Source

Public dataset: TrashNet (Thung and Yang), one waste item per photo

Size

2527 images in 6 classes, resized to 224x224 for EfficientNet-B0

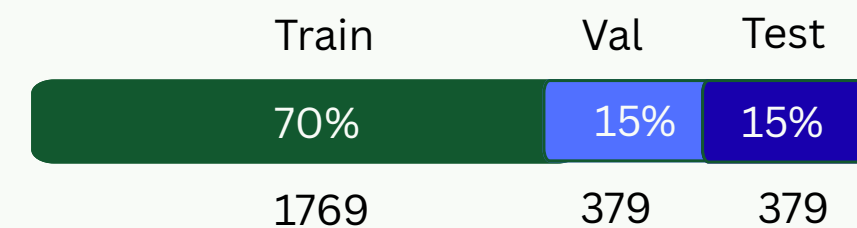
Labels

One label per image: Cardboard, Glass, Metal, Paper, Plastic, or Trash

Images per class



Train / validation / test split



Stratified split. The test stays untouched until final evaluation



Training only: Flips, rotation, and brightness/color jitter

Cleaning: Check for unreadable images and duplicates before splitting

Extra: A few of our own photos to test real-world performance

Success Metrics

We will evaluate accuracy, speed, and classification quality against measurable project goals.



PRIMARY METRIC

$\geq 90\%$

Test Accuracy

Correctly classify waste items into 6 categories on the test set.



SECONDARY METRIC

$\leq 1 \text{ sec}$

Inference Time

Average time to classify one image on Google Colab.



ADDITIONAL METRIC

≥ 0.85

Macro F1-score

Measure overall classification performance across all classes.



Evaluation Method

- Test on a held-out TrashNet test set
- Use a confusion matrix to analyze per-class performance
- Report accuracy, precision, recall, and Macro F1-score



Milestone Plan

A phased plan keeps the project moving from a working demo to a measurable final application.



8 Risks & Resources

 Smart Recycling

We identified the main risks and prepared practical backup plans using free, accessible tools.

Risks & Plan B

1 Similar-looking objects

Plastic, glass, or paper items can look visually similar.

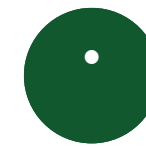
Plan B: Use training augmentation and tune the model using validation data.

2 Dataset / real-world variation

Lighting, angles, and backgrounds may differ from TrashNet.

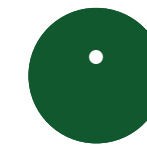
Plan B: Add sample images and fine-tune further with augmentation.

Resources & Cost



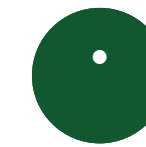
PyTorch

Deep learning
framework



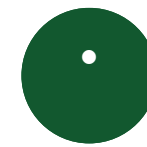
Google Colab

Free GPU
compute



TrashNet

2,527 images
6 classes (Thung
and Yang)



GitHub

Version control
+ project repo

Estimated Cost

\$0

All core tools, dataset, and compute are free to use.

Goal:

Keep the project technically realistic, affordable, and finishable within the term.

● References ●

- PyTorch. “Efficientnet_B0 — Torchvision 0.26 Documentation.” Pytorch.Org, 2024, docs.pytorch.org/vision/stable/models/generated/torchvision.models.efficientnet_b0.html. Accessed 5 Oct. 2026.
- Thung, Gary, and Mindy Yang. “Garythung/Trashnet.” GitHub, 2 Dec. 2020, github.com/garythung/trashnet. Accessed 5 Oct. 2026.

